

Design of an Elevated CPW for High-Speed Digital Circuit Applications

*Yoon-Jae Hwang, *Jong-Gwan Yook, **Yong-Jun Kim, *Han-Kyu Park

*Dept. of Electronical and Electronics Engineering, Yonsei Univ,
134, shinchon-Dong, Seodaemun-Gu, Seoul, Korea, 120-749

**Dept. of Mechanical Engineering, Yonsei Univ
Tel : +82-2-2123-4618, Email : jgyook@yonsei.ac.kr

Abstract - In designing and manufacturing of high speed digital circuits it is important to develop a transmission line having characteristics of a higher phase velocity and a wide operating frequency. Recently, the use of an elevated coplanar waveguide (ECPW) has been proposed to achieve such characteristics[1]. A full-wave analysis of such a multi-layer ECPW using 2D-FDTD will be presented here. Various heights and widths of conductor and slot are considered to give design guidelines.

Introduction

Today monolithic microwave integrated circuits (MMICs) with designs based on microstrip line and coplanar waveguide (CPW) are applied in several fields. Due to its ease of mounting devices, CPW is a fundamental and important element for MMICs. But when CPW operates in high frequency it shows dispersive characteristics caused by air-dielectric interface. Hereby it is needed to elevate signal line in the air to diminish its discontinuity[1,2,3]. The 2D-FDTD method has been developed as a versatile method for the simulation of open and closed boundary structures with arbitrary cross section. Using 2D-FDTD various heights and conductor and slot widths are considered to give design guidelines for a proposed ECPW as shown in Fig. 1.

Validation

Before 2D-FDTD is applied to the proposed ECPW, we have established the validity of 2D-FDTD by comparing published data. Tested microstrip line is shown in Fig. 2 and the resonant frequency is displayed in Fig. 3. Generally, the resonant frequency corresponds to a position at which magnitude has a peak and phase has 180° phase change in frequency domain EM field data. We can find that the resonant frequency of the tested microstrip line is 5.565 GHz and then also find effective dielectric constant as 6.613. This result agrees well with the reference[3] within 1.3%. Also we used a perfect electric conductor (PEC) as an outer boundary condition. As shown in Fig. 4, it is found that the resonant frequency of a dominant mode is consistent irrespective to whether open or closed outer boundary conditions are used.

Numerical Results

Case 1) ABC (1st Mur's ABC) outer boundary condition

As can be seen from Fig. 5, the resonant frequency of the ECPW is higher than that of a normal CPW. As expected in [1], effective dielectric constant of the ECPW is lower than that of a normal CPW, and then that results in a higher phase velocity. This is because the air-dielectric discontinuity effect is diminished as elevating signal line from dielectric substrate. When the width of signal line decreases, the phase velocity increases. Similarly, as the width of slot increases the phase velocity

decreases. That is, narrow signal line, slot as well as elevation cause the higher phase velocity. Also the use of a $10\ \mu\text{m}$ thick polyimide buffer layer attributes to a higher phase velocity. Table I illustrates the above results.

Case 2) PEC outer boundary condition

Fig. 6 suggests that a shielded ECPW could guarantee a wide operating frequency by suppressing the second order mode near 50 GHz. Regardless of shielding, it is observed that the ECPW has a higher phase velocity as summarized in Table II. Comparing Fig. 6 with Fig. 5, it is found that the resonant frequency of a dominant mode is independent whether open or closed outer boundary conditions are used

Conclusions

An ECPW is a structure that could support transmission of EM waves and signals through near-TEM waveguide by separating the signal line far from dielectric substrate as possible. As can be recognized from computational results using 2D-FDTD, when the ECPW is adapted to high speed digital circuits, better performance is expected in that the ECPW has a higher phase velocity and a wide operating frequency. Moreover Multi-layer geometry and narrow signal line and slot width could provide additional advantages of this ECPW configuration.

Acknowledgement

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References

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TABLE I

Comparison of The Line Parameters for Different Cases (ABC) * elevated case

	Resonant frequency (GHz)	Effective dielectric constant	Phase velocity ($\times 10^8$ m/s)
Normal ($h=0\ \mu\text{m}$)	21.45	4.95	1.39
*Elevated ($h=10\ \mu\text{m}$)	28.15	2.88	1.77
*Without Multi-layer	26.82	3.17	1.69
*Wide signal line ($w=200\ \mu\text{m}$)	25.48	3.51	1.60
*Wide slot ($s=150\ \mu\text{m}$)	26.81	3.17	1.69

TABLE II
Comparison Between Normal and Elevated Case (PEC)

	Resonant frequency (GHz)	Effective dielectric constant	Phase velocity ($\times 10^8$ m/s)
Normal ($h=0 \mu m$)	23.88	4.00	1.50
Elevated ($h=10 \mu m$)	31.65	2.28	1.99

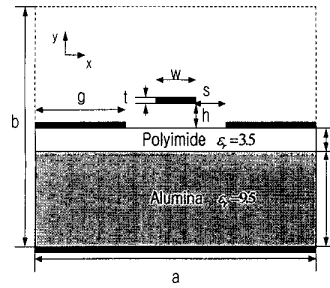


Fig. 1. Cross-section of proposed ECPW
($a=1300 \mu m$, $b=400 \mu m$, $d1=100 \mu m$, $d2=10 \mu m$,
 $w=100 \mu m$, $g=500 \mu m$, $s=100 \mu m$, $h=10 \mu m$, $t=0 \mu m$)

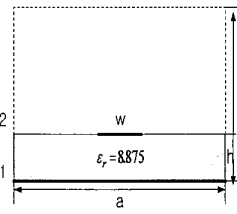


Fig. 2. Tested microstrip line
($a=b=12.7 mm$, $w=h=1.27 mm$)

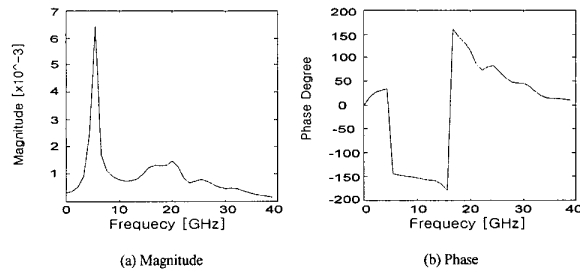


Fig. 3. The spectrum of tested microstrip line

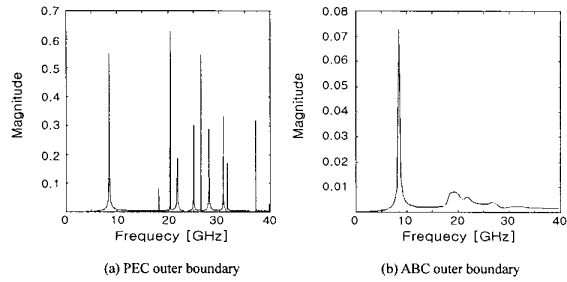


Fig. 4. The resonant frequency for different boundary conditions

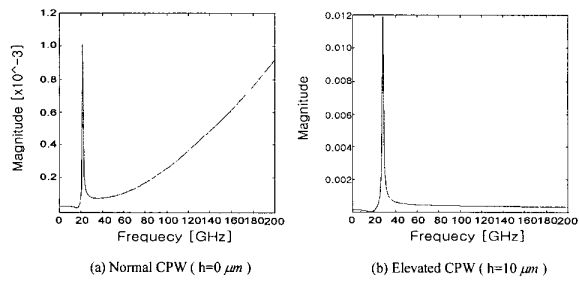


Fig. 5. The resonant frequency of normal and elevated CPW

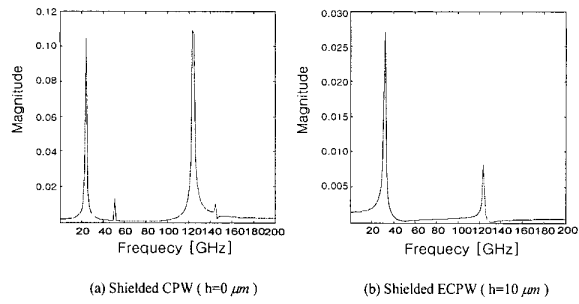


Fig. 6. The resonant frequency of shielded normal and elevated CPW